SOURCE CODE

```
// PIR Motion Detection using Arduino UNO

int pirPin = A0;   // PIR output connected to A0
int motionState = 0;

void setup() {
  pinMode(pirPin, INPUT);
  Serial.begin(9600);
}

void loop() {
  motionState = digitalRead(pirPin);

  if (motionState == HIGH) {
    Serial.println("Motion Detected!");
  } else {
    Serial.println("No Motion");
  }

  delay(500);
}
```

EXPLANATION

A PIR (Passive Infrared) Sensor is an electronic device used to detect the movement of people or objects by sensing changes in infrared (heat) radiation. Human bodies naturally emit infrared energy, and when a person moves within the sensor's detection range, the PIR sensor identifies the change and generates an output signal.

In this project, the PIR sensor is connected to an Arduino Uno microcontroller. The Arduino continuously monitors the sensor's output using the digitalRead() function. Under normal conditions, when no motion is present, the sensor output remains LOW. When motion is detected, the sensor output changes to HIGH.

The Arduino processes this signal and displays the status on the Serial Monitor. If motion is detected, the message "Motion Detected!" is displayed. If no motion is detected, the message "No Motion" is shown. This allows the user to monitor activity in the sensor's surroundings in real time.

A short delay is included in the program to ensure stable and reliable sensor readings. The system can be further expanded by connecting devices such as LEDs, buzzers, or alarms to provide visual or audible alerts whenever motion is detected.

CIRCUIT DIAGRAM